

COURSE DESCRIPTION			
Program	Food Processing Technology		
Course Title	Food Processing Technology lab		
Course Code	2429	Semester	II
Type of Course	Lab	Contact Hours	45
Teaching Scheme (L: T:P:R)	0:0:3:0	Credits	1.5
CIE Marks	60	ESE Marks	40

Introduction

This course is designed to provide students with practical and theoretical knowledge of food processing and preservation techniques across fruits, vegetables, dairy, and other food products. It emphasizes understanding and applying standardized operations to develop safe, stable, and high-quality food products. The course integrates hands-on experience with quality evaluation methods, enabling students to assess the physicochemical, sensory, and storage characteristics of processed foods. This knowledge equips students with the skills necessary for innovation, product development, and ensuring compliance with food safety and quality standards in the food industry.

Course Objectives

1	Apply and assess standardised fruit processing and preservation techniques for quality and safe beverages and preserves.
2	Demonstrate vegetable processing operations to develop stable products with desirable quality attributes.
3	Apply dairy and plant-based protein processing methods for preparing and evaluating traditional and value-added products.
4	Utilise food quality evaluation techniques to assess sensory, physicochemical, and shelf-life characteristics of processed foods.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic knowledge of food processing techniques	1421	Fundamentals of food processing technology	I

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Apply and evaluate standardised fruit processing and preservation techniques for the development of beverages and preserves, with emphasis on product quality, stability, and safety standards.	Understand
CO2	Demonstrate the application of vegetable processing operations such as curing, concentration, and minimal processing to develop stable vegetable products with desirable quality attributes.	Understand
CO3	Apply appropriate dairy and plant-based protein processing techniques, including coagulation, heat treatment, packaging, and storage, for the preparation and quality evaluation of traditional and value-added milk products.	Understand
CO4	Apply food quality evaluation techniques to assess sensory, physicochemical, and shelf-life characteristics of milk and processed food products under different storage and processing conditions.	Understand

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	2	-	-	-	-	2	2	-	-	-	2
CO2	2	-	-	-	-	2	2	-	-	-	2
CO3	2	-	-	-	-	2	2	-	-	-	2
CO4	2	-	-	-	-	2	2	-	-	-	2
Total	8	-	-	-	-	8	8	-	-	-	8
Correlation Level	2	-	-	-	-	2	2	-	-	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-"

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
0	0	3	0	1.5	45	1.5				60	40	100	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Detailed Syllabus

Expt No.	Experiment name	Practical Outcome	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
1	Standardisation of formulation and processing parameters for Fruit syrup	Standardise formulations and processing parameters for the manufacture of fruit syrup and evaluate product consistency and quality.	CO1	Understand	3
2	Preparation and quality evaluation of Marmalade	Prepare marmalade using appropriate processing techniques and assess its quality attributes such as TSS, acidity, and sensory characteristics.	CO1	Understand	3
3	Preparation, quality evaluation of Squash	Formulate and process fruit squash and evaluate its physico-chemical and sensory quality.	CO1	Understand	3
4	Preparation, packaging and storage of RTS	Prepare, package, and store RTS beverages, and analyze the effect of packaging and storage conditions on product stability.	CO1	Understand	3
Module II					
5	Preparation and packaging of Vegetable Pickle using different curing methods	Prepare and package vegetable pickle using different curing methods and evaluate their effect on product quality and stability.	CO2	Understand	3
6	Preparation of tomato puree	Process tomatoes into puree and paste using appropriate concentration and heat treatment techniques.	CO2	Understand	3
7	Preparation of tomato paste	Apply minimal processing techniques to vegetables to maintain freshness, safety, and sensory quality.	CO2	Understand	3
8	Minimal processing of vegetables	Assess the quality attributes of vegetable-based products during processing and storage.	CO2	Understand	3
Module III					
9	Preparation, packaging and storage of Khoa	Prepare, package, and store khoa, and evaluate changes in quality during storage.	CO3	Understand	3
10	Preparation, packaging and storage of rasagola	Prepare rasagola using standardized processing steps and assess product quality.	CO3	Understand	3
11	Processing of tofu	Process tofu using suitable coagulation techniques and evaluate its texture and quality.	CO3	Understand	3
12	Effect of Coagulants on Paneer Quality	Analyze the effect of different coagulants on paneer quality, including yield, texture, and sensory attributes.	CO3	Understand	3
Module IV					
13	To study the sensory and other quality parameters of the milk	Evaluate the sensory and physicochemical quality parameters of milk, including appearance, flavour, pH, and acidity.	CO4	Understand	3
14	To study the effect of temperature on the keeping quality of milk	Analyse the effect of temperature on the keeping quality of milk by monitoring changes in quality attributes during storage.	CO4	Understand	3
15	To evaluate the quality of a processed food product during storage using physicochemical parameters.	Assess the quality of processed food products during storage using relevant physicochemical parameters.	CO4	Understand	3
16	To compare the sensory and shelf-life characteristics of homemade and commercially processed food products.	Compare sensory quality and shelf-life characteristics of homemade and commercially processed food products.	CO4	Understand	3

Open-ended experiment

Objective:

To encourage creativity, innovation, and application of knowledge by allowing students to design and perform an experiment beyond the prescribed list, using the concepts learned during the course.

Description:

At the end of the laboratory course, each student (or group of students) shall undertake one open-ended experiment. In this activity, students are expected to conceive, design, and implement an experiment of their own choice related to the course outcomes. The aim is to apply theoretical knowledge and laboratory skills to a new or practical problem not explicitly covered in the regular experiments.

Students may choose to:

Modify or extend an existing experiment to explore additional parameters or behaviors.

Integrate multiple concepts from the syllabus into a combined application.

Design a mini-setup or prototype to demonstrate a principle or solve a simple engineering problem.

Perform a comparative study or performance analysis of a circuit/system/component.

Student Activities for Self-Learning

Week	Self-Learning description	Hours
Module I		
1	Prepare a fruit jam or jelly using a standardized recipe and record observations on consistency and flavor.	1.5
2	Conduct a small-scale beverage processing (juice or squash) and evaluate shelf stability over a week.	1.5
3	Compare preservation methods (sugar, refrigeration, pasteurization) for a chosen fruit and note differences in quality.	1.5
4	Research and present a case study on commercial fruit processing techniques and quality standards.	1.5
Module II		
5	Prepare minimally processed vegetables (blanched, frozen, or vacuum-packed) and observe changes in color, texture, and flavor.	1.5
6	Perform a simple curing or pickling experiment and assess the effect of salt/acidity on shelf life.	1.5
7	Study concentration techniques (like dehydration) on a chosen vegetable and evaluate retention of nutrients and color.	1.5
8	Create a flowchart of industrial vegetable processing operations for a selected product.	1.5
Module III		
9	Prepare paneer, yogurt, or a plant-based milk product and evaluate coagulation and texture.	1.5
10	Compare heat treatment methods (pasteurization, boiling) on milk and record differences in microbial safety and shelf life.	1.5
11	Conduct a sensory evaluation of a processed dairy or protein product for taste, texture, and appearance.	1.5
12	Research innovative plant-based protein products and summarize processing techniques.	1.5
Module IV		
13	Perform a simple sensory evaluation (color, taste, aroma) of a fruit, vegetable, or milk product.	1.5
14	Measure physicochemical parameters (pH, acidity, viscosity) for a processed food item.	1.5
15	Prepare a report comparing lab results with standard quality specifications from industry guidelines.	1.5
Total Self-Learning hours		22.5

Learning Resources

TEXT BOOK			
Sl. No.	Title	Author	Publication
1	Food Processing and Preservation	P.J. Fellows	CRC Press, 2020
2	Principles of Food Science	N. Potter & J. Hotchkiss	Springer, 2019
3	Fruit and Vegetable Processing	A. K. J. Reddy	PHI Learning, 2018
4	Dairy Technology	S. K. Hui	Wiley-Blackwell, 2021
5	Food Quality Assurance	Norma B. Chadwick	Springer, 2017

REFERENCE			
Sl. No.	Title	Author	Publication
1	Handbook of Food Science, Technology, and Engineering	Y. H. Hui	CRC Press, 2017

2	Plant Proteins: Processing and Product Development	Y. Chen & S. Ma	Academic Press, 2018
3	Sensory Evaluation of Food	Morten Meilgaard	Springer, 2020

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.fao.org/food-processing/en/	I
2	https://www.ars.usda.gov/	II
3	https://www.sciencedirect.com/topics/food-science	All

Major Equipment/Instruments/Components required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
1	Hot plate/stove	Electric hot plate (single or double burner) or gas stove (for traditional setups)	1
2	refractometer	Erna Hand Refractometer - 1)Brix 28-60' 2)Brix 58-90'	2 each
3	Blender	Laboratory-grade or high-quality domestic blender	1 set
4	thermometer	Mercury / Alcohol Laboratory Thermometer	4
5	refrigerator	domestic refrigerator - 0°C to 10°C (fresh), -18°C to -20°C (frozen storage)	1
6	weighing balance	Accuraxy 0.1 G Max 6Kg Make Saffron small precise value Analytical balance 0.001 g (1 mg)	1 each
7	spectrophotometer	UV-Visible / Visible Spectrophotometer: 200-700 nm wavelength	1
8	colorimeter	Lovibond or digital colorimeter - Red, Green, Blue / or selectable wavelength 400-700 nm	1
9	viscometer	Brookfield or rotational viscometer	1

Major Software required for a batch of 30 students

Sl. No.	Particulars	Specifications	Quantity
	NIL		

Assessment Methodology - Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE
		Continuous evaluation	Practical test (first half of experiments)	Practical test (second half of experiments)	Open-ended experiment	Practical
Duration			3 hours	3 hours		3 hours
Exam mark		50	40	40	50	40
Converted to		30	15	15	10	
Marks	5	30	15		10	40
Total marks	60					40
Tentative schedule	End of semester	Every lab slot	7th week	14th week	15th week	

Scheme of Evaluation - Practical (Continuous Internal Evaluation)

Sl. No.	Criteria	Description	Marks
1	Preparation & Punctuality	Attendance, timely submission of record, pre-lab preparation, understanding of experiment objectives and theory.	10
2	Experimental Setup & Procedure	Ability to identify components/equipment, correct circuit connections/setup, systematic and safe execution of the procedure.	10
3	Observation & Data Recording	Accuracy and neatness of readings, correct use of units, tabulation, and systematic recording of results.	5
4	Analysis & Result Interpretation	Proper calculations, graphical representation, result verification, discussion, and logical conclusion.	10
5	Viva Voce / Concept Understanding	Clarity of theoretical concepts, explanation of procedure, ability to answer questions related to the experiment.	10
6	Workmanship, Discipline & Teamwork	Laboratory discipline, care of instruments, teamwork, attitude, and safety practices.	5

Total	50
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Scheme of Evaluation - Practical Test & End Semester Evaluation

Category	Things to evaluate	Marks
Write-up / Procedure (before experiment)	Aim, circuit diagram/flowchart, stepwise procedure.	10
Experiment Setup & Execution	Correct circuit connections / coding / setup. Handling of instruments/software. Accuracy of execution.	10
Observation & Result / Output	Proper tabulation of readings. Calculations/graphs/program output. Correctness of result / expected outcome.	8
Viva Voce	Conceptual understanding of experiment. Related theoretical knowledge.	8
Record	Completion & neatness of practical record. Proper certification by faculty.	4
Total		40

Scheme of Evaluation - Open-ended experiment

Category	Things to evaluate	Marks
Originality and Relevance of the Idea	Assesses how creative, innovative, and relevant the chosen experiment is to the course objectives.	10
Clarity of Objectives and Experimental Plan	Evaluates how clearly the student defines the aim, scope, and systematic plan of the experiment.	10
Correctness of Execution and Data Recording	Checks the accuracy, safety, and neatness in performing the experiment and recording observations.	10
Analysis, Interpretation, and Presentation of Results	Measures the student's ability to analyze data, interpret outcomes, and present results logically.	10
Teamwork and Innovation Demonstrated	Judges the level of collaboration, initiative, and creativity shown during the experiment.	10
Total		50